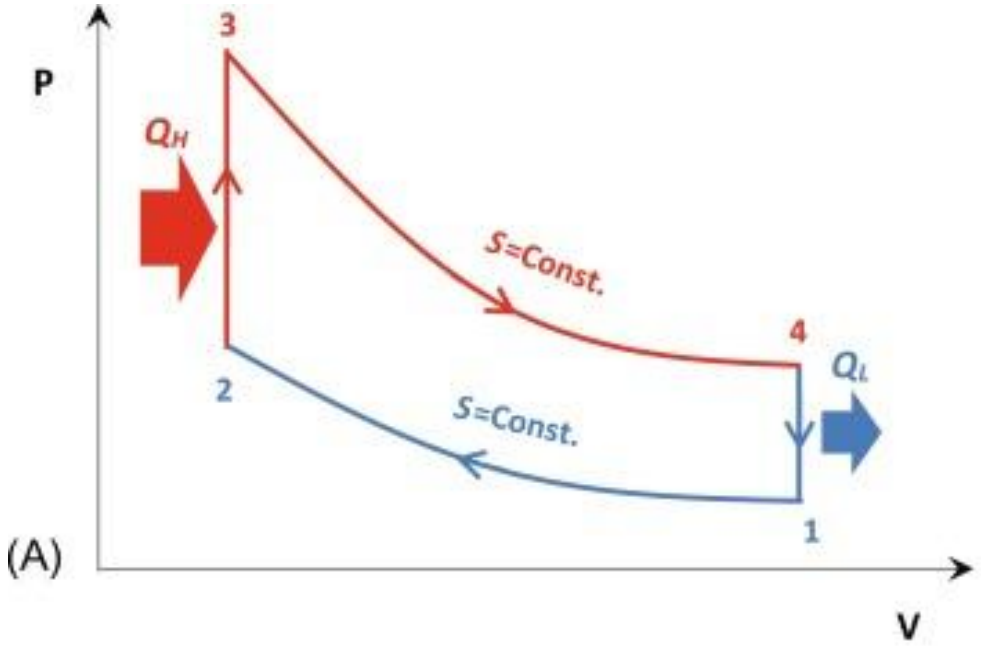
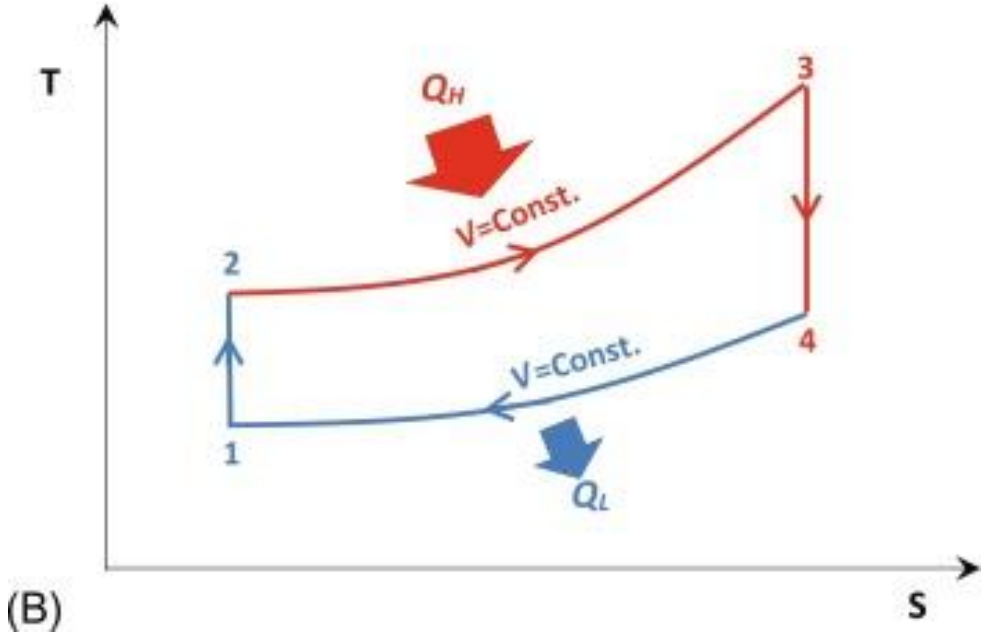
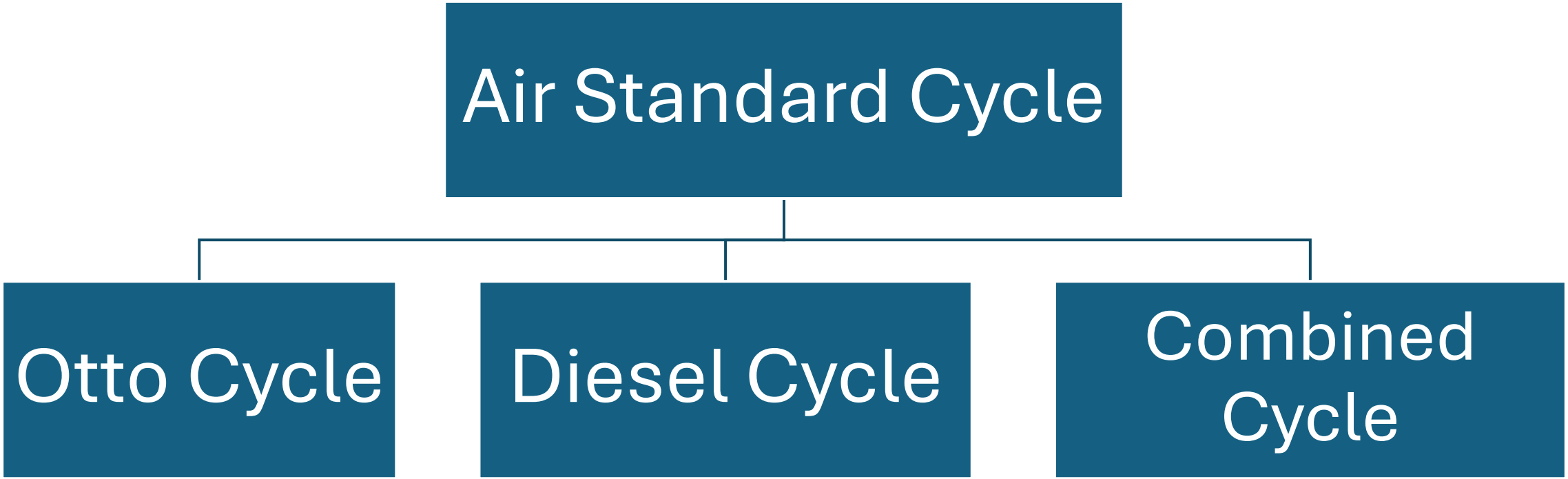




Chapter 2

Air Standard Cycles


Air Standard Cycle
Otto Cycle
Diesel Cycle
Combined Cycle

Replace the real engine cycle by an ideal closed cycle using air

Main air-standard assumptions

1. **The working fluid is air throughout the cycle.**
   No actual fuel-air combustion products are modeled.
2. **Air behaves as an ideal gas.**

$$PV = mRT$$

3. **The cycle is closed.**
   The same air circulates continuously. There is no real intake or exhaust process.
4. **Combustion is replaced by external heat addition.**

$$\text{Combustion} \rightarrow Q_{in}$$

5. **Exhaust is replaced by heat rejection.**

$$\text{Exhaust gases} \rightarrow Q_{out}$$

6. **Compression and expansion are assumed reversible and adiabatic**, therefore:

isentropic

7. In the basic air-standard model, changes in kinetic and potential energy are neglected.

$R_{\text{air}} \approx 0.287$ kJ/kg·K ≈ 287 J/kg·K

Air-standard cycle
is an idealized thermodynamic model used to analyze internal-combustion engines without dealing with the full complexity of real combustion, exhaust, intake, chemical reactions, heat losses, etc.

$$Pv = RT$$

$$v = {}^{V}/_{m}$$

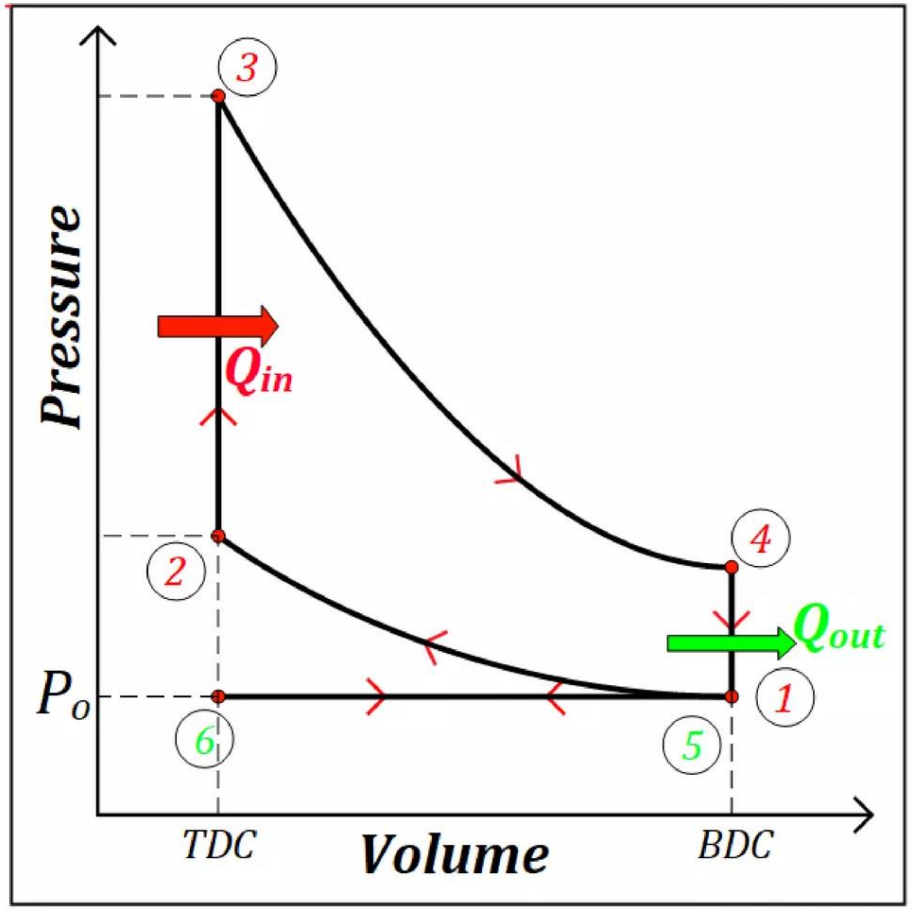
$V =$ total volume

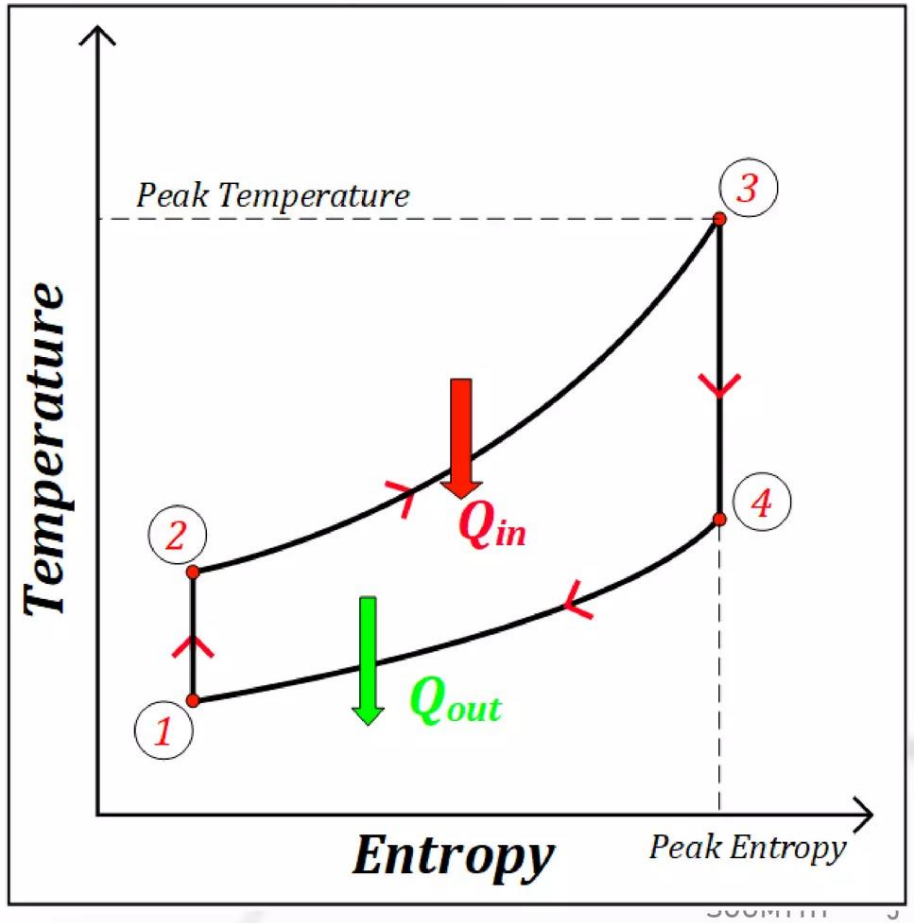
$v =$ specific volume $=$ volume per kg

Process	What stays constant?	Main relation for ideal gas	Typical ICE example
Isothermal	Temperature T	PV = Constant	Not normally an ideal engine-cycle process
Isobaric	Pressure P	V/T =Constant	Heat addition in ideal Diesel cycle
Isochoric / Isovolumetric	Volume V	P/T =Constant	Heat addition in ideal Otto cycle
Isentropic	Entropy s	PV^{γ} =Constant	Ideal compression and expansion
Adiabatic	No heat transfer, $Q = 0$	Depends on reversibility	Approximation for fast compression/expansion
Polytropic	No single property necessarily constant	PV^{n} =Constant	Real compression/expansion approximation
Isenthalpic	Enthalpy h	h =Constant	Throttling valves
Isentropic + adiabatic reversible	s =, $Q = 0$	PV^{γ} =Constant	Ideal compressor/turbine/piston process

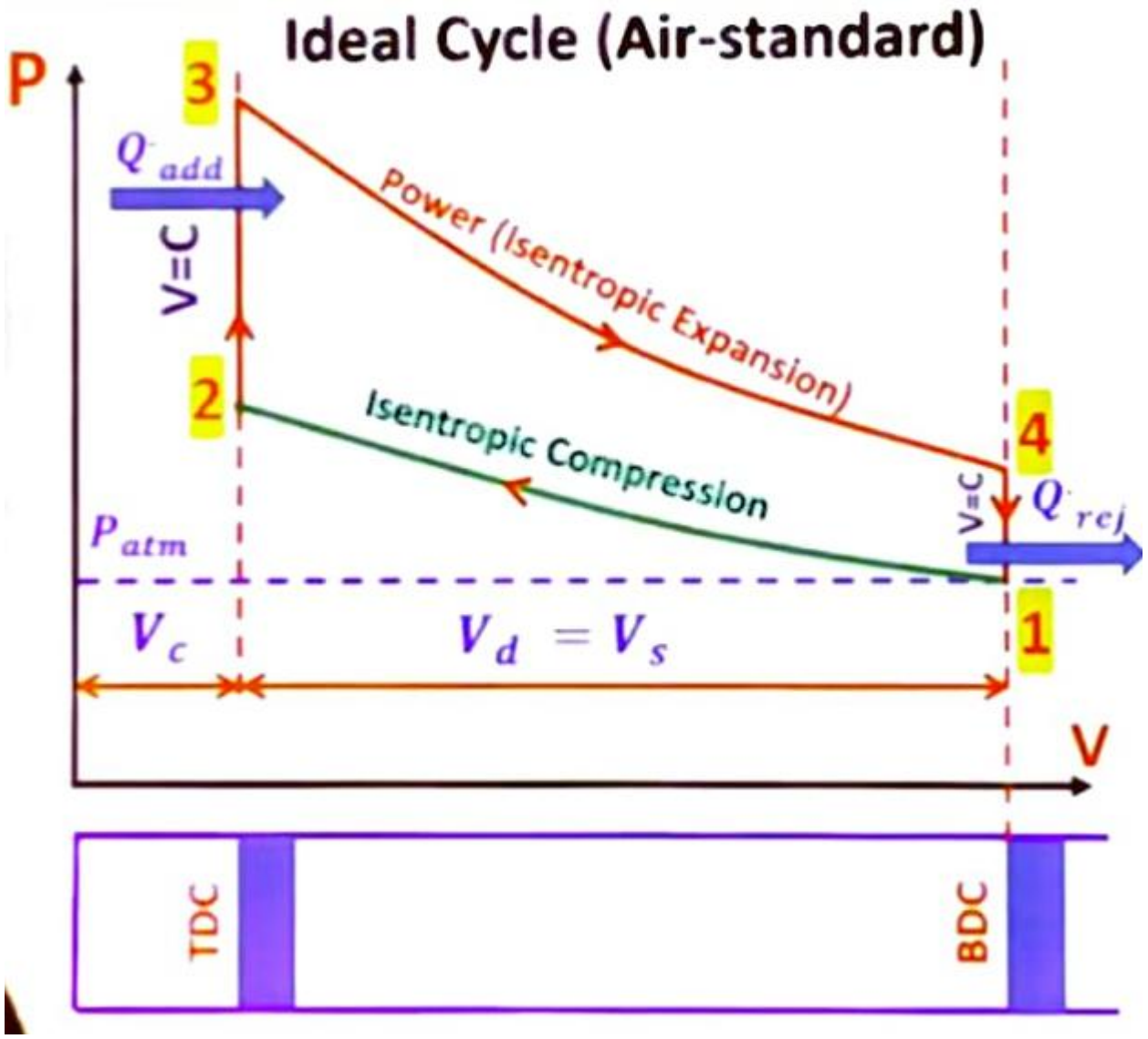
Gasoline cycle

Otto Cycle





1 — 2	Isentropic Compression
2 — 3	Constant Volume Combustion
3 — 4	Isentropic Expansion
4 — 1	Constant Volume Heat Rejection



$$V_d = \frac{\pi}{4} D^2 L$$

Compression ratio

$$r_c = \frac{V_1}{V_2} = \frac{V_4}{V_3}$$

Pressure ratio

$$r_P = \frac{P_3}{P_2}$$

Heat added

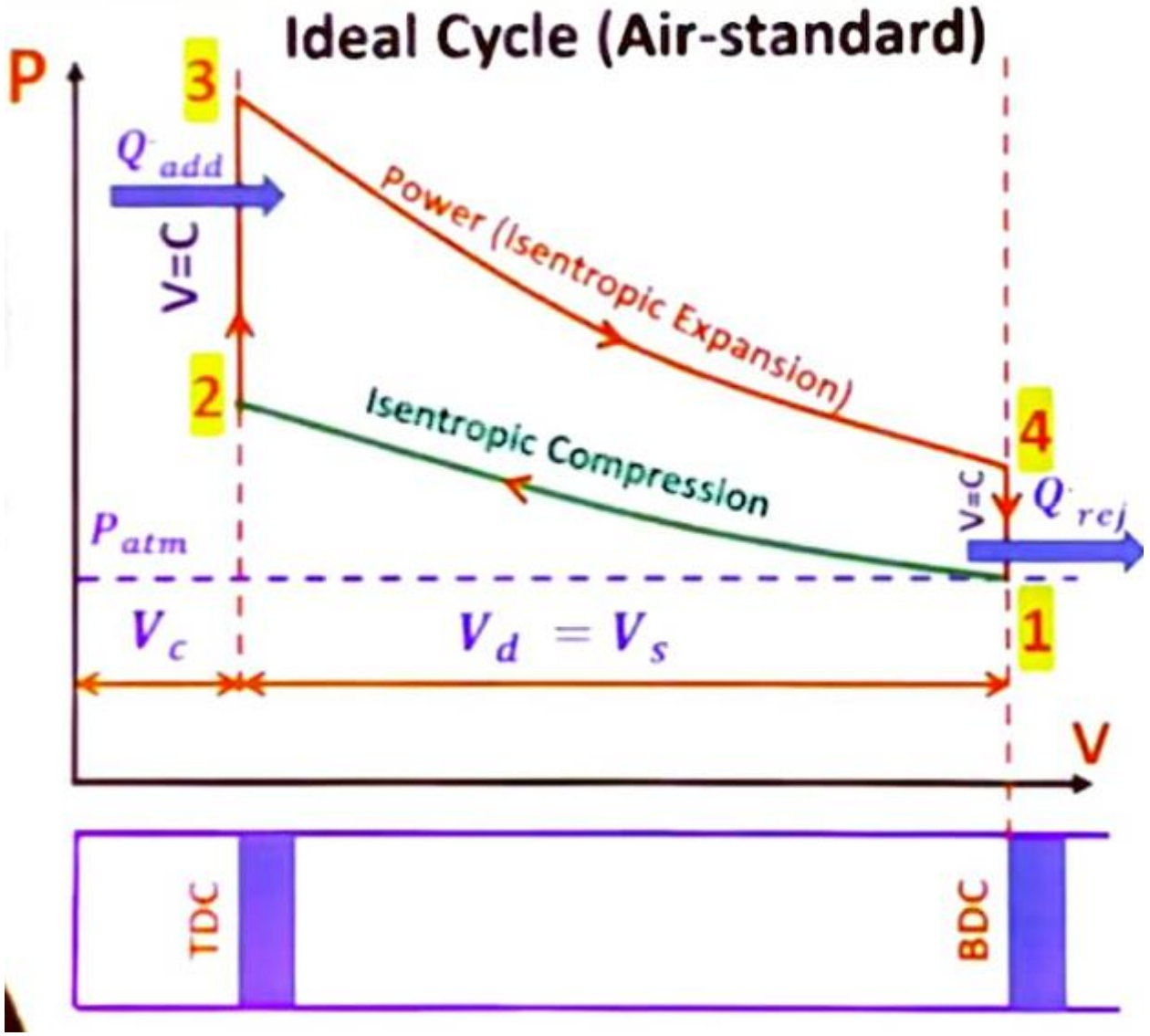
$$Q_{add} = m_m c_v (T_3 - T_2)$$

$$q_{add} = c_v (T_3 - T_2)$$

Heat rejected

$$Q_{rej} = m_m c_v (T_4 - T_1)$$

$$q_{rej} = c_v (T_4 - T_1)$$



Compression work

$$W_{comp} = m_m c_v (T_2 - T_1)$$

$$w_{comp} = c_v (T_2 - T_1)$$

Expansion work

$$W_{exp} = m_m c_v (T_3 - T_4)$$

$$w_{exp} = c_v (T_3 - T_4)$$

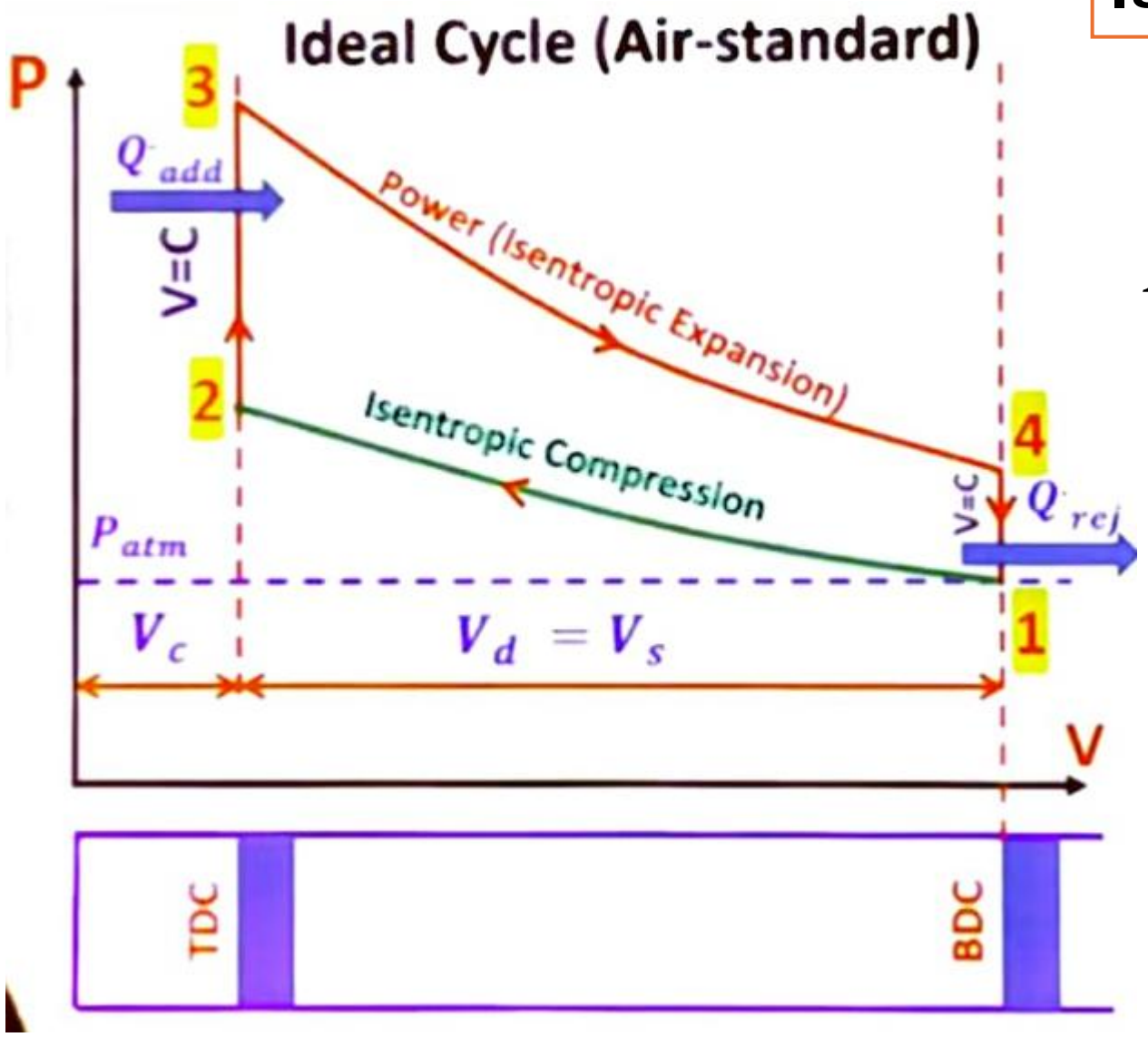
Net work

$$W_{net} = W_{comp} - W_{exp}$$

$$W_{net} = Q_{add} - Q_{rej}$$

$$w_{net} = q_{add} - q_{rej}$$

Otto Cycle analysis



Process 1-2
Isentropic compression

$$P_1 v_1 = RT_1$$

$$r_c = \frac{v_1}{v_2}$$

$$P_1, T_1, v_1, r_c \text{--- } P_2, T_2, v_2$$

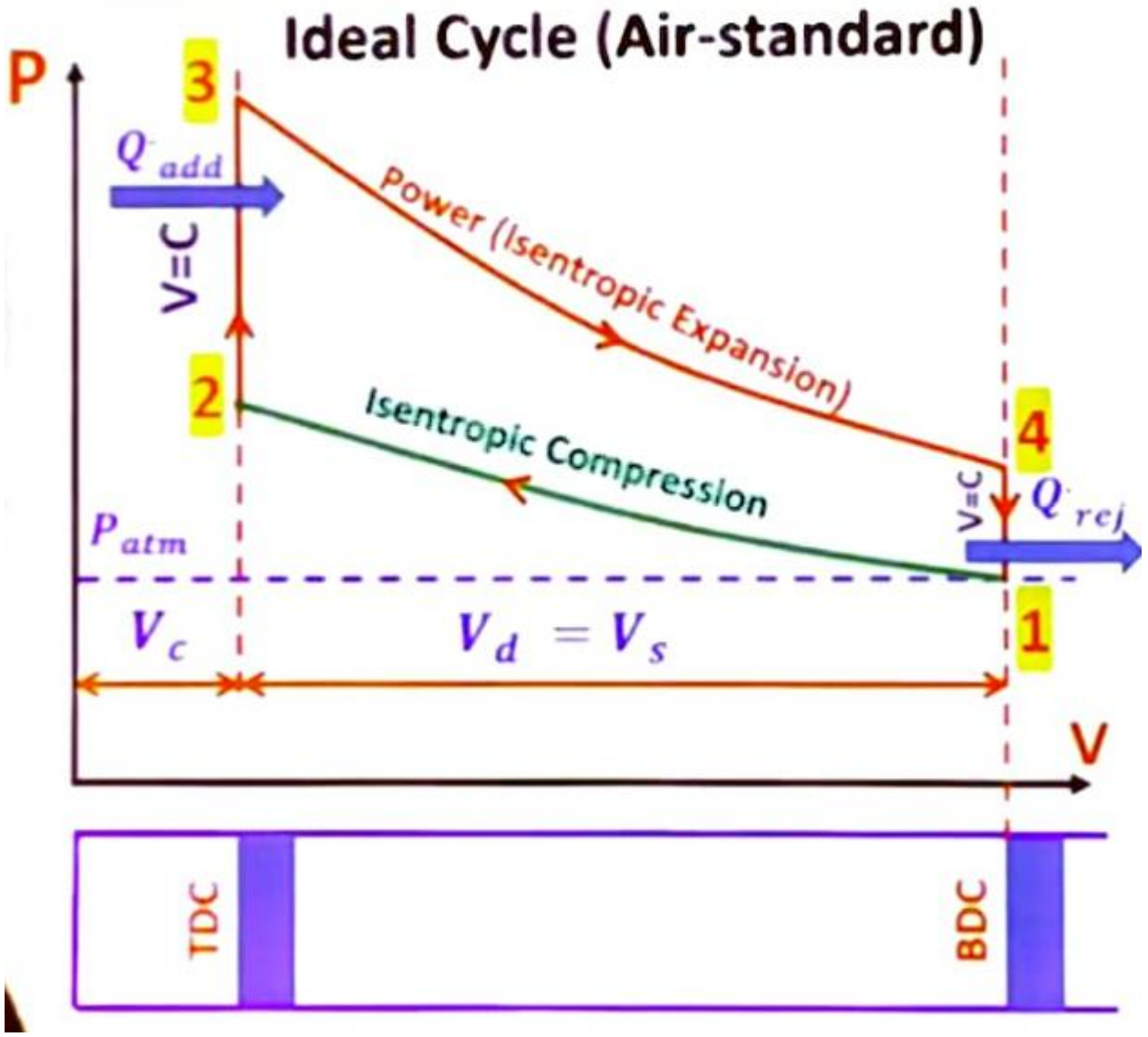
$\gamma = 1.4$ for air

$$\frac{T_2}{T_1} = \left(\frac{v_1}{v_2}\right)^{\gamma-1}$$

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$$

$$\frac{P_2}{P_1} = \left(\frac{v_1}{v_2}\right)^{\gamma}$$

Otto Cycle analysis



Process 2-3
Constant volume heat addition

$$V = const$$

$$v_2 = v_3$$

$$P_3 v_3 = R T_3$$

3 Possible solutions

T_{max}
r_P
Q_{fuel} or A/F

$$r_P = \frac{P_3}{P_2} = \frac{T_2}{T_3}$$

Process 2-3
Constant volume heat addition

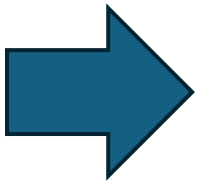
$$Q_{fuel} = m_f HV\ \eta_c$$

$$Q_{add} = m_m c_v (T_3 - T_2)$$

LHV Or HHV.... fuel heating value or calorific value CV

η_c *combustion efficiency*

$$m_f HV\ \eta_c = m_{mix} c_v (T_3 - T_2)$$

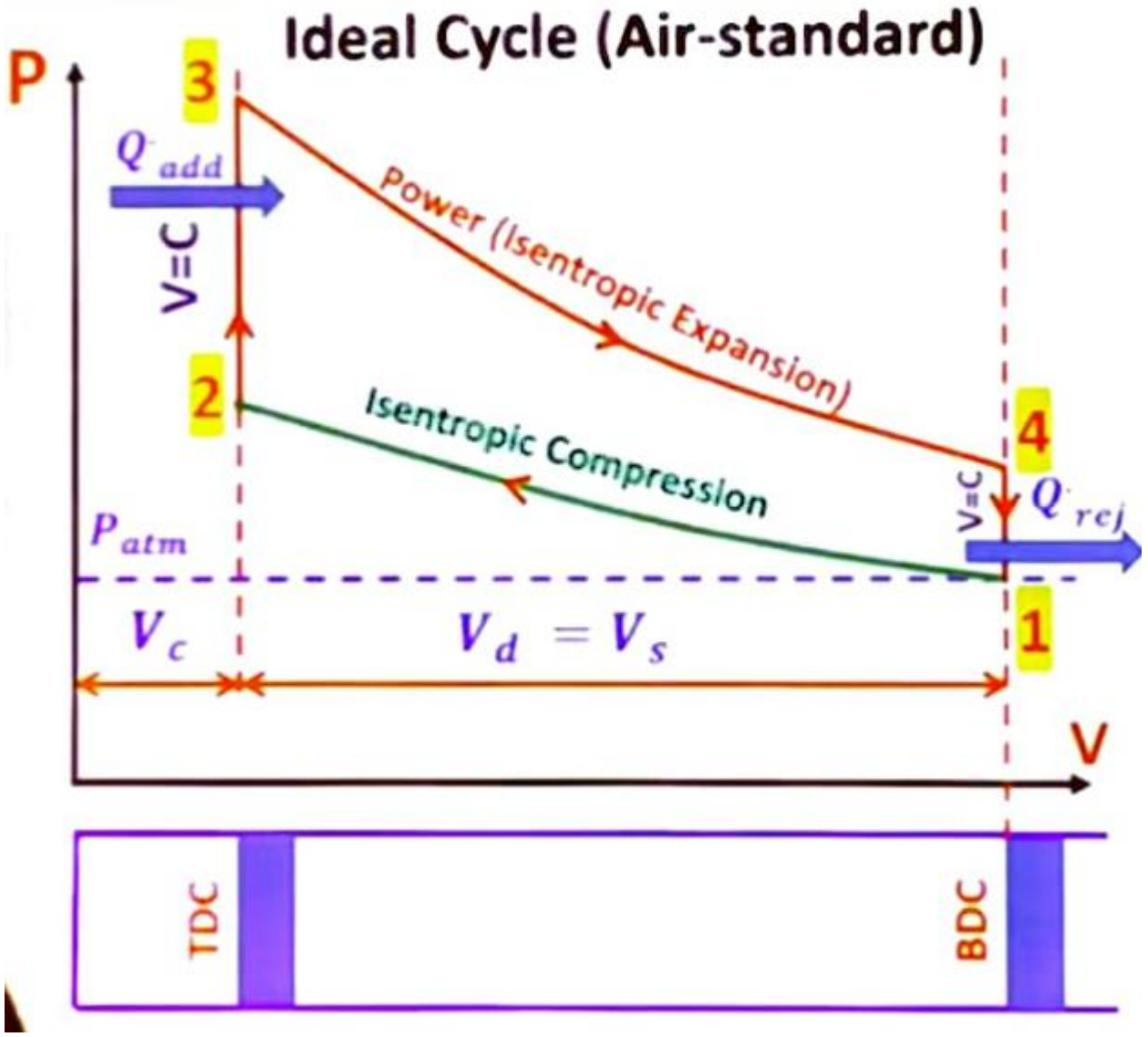


T_3 & P_3

$$m_{fuel} HV\ \eta_c = (m_{air} + m_{fuel}) c_v (T_3 - T_2)$$

$$HV\ \eta_c = \left(\frac{A}{F} + 1\right) c_v (T_3 - T_2)$$

Otto Cycle analysis



Process 3-4
Isentropic Expansion

$$\frac{T_3}{T_4} = \left(\frac{v_4}{v_3}\right)^{\gamma-1}$$

$$\frac{T_3}{T_4} = \left(\frac{P_3}{P_4}\right)^{\frac{\gamma-1}{\gamma}}$$

$$\frac{P_3}{P_4} = \left(\frac{v_4}{v_3}\right)^{\gamma}$$

$$r_c = \frac{v_1}{v_2} = \frac{v_4}{v_3}$$

Otto Cycle Summary

$$Q_{add} = m_m c_v (T_3 - T_2)$$

$$q_{add} = c_v (T_3 - T_2)$$

$$Q_{rej} = m_m c_v (T_4 - T_1)$$

$$q_{rej} = c_v (T_4 - T_1)$$

$$W_{comp} = m_m c_v (T_2 - T_1)$$

$$w_{comp} = c_v (T_2 - T_1)$$

$$W_{exp} = m_m c_v (T_3 - T_4)$$

$$w_{exp} = c_v (T_3 - T_4)$$

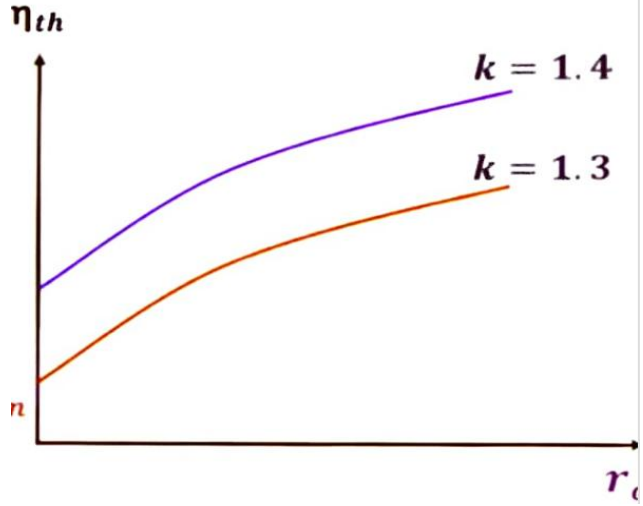
$$W_{net} = Q_{add} - Q_{rej}$$

$$w_{net} = q_{add} - q_{rej}$$

Thermal efficiency = $\eta_{th} = \frac{W_{net}}{Q_{add}} = \frac{w_{net}}{q_{add}}$

$$\eta_{th} = \frac{Q_{add} - Q_{rej}}{Q_{add}} = 1 - \frac{Q_{rej}}{Q_{add}}$$

$$\eta_{th} = 1 - \frac{1}{r_c^{\gamma - 1}}$$



Mean effective pressure

$$P_m = \frac{W_{net}}{V_d}$$

$$W_{net} = m_m c_v[(T_3 - T_2) - (T_4 - T_1)]$$

$$T_2 = T_1 r_c^{\gamma-1}$$

$$T_3 = r_p T_1 r_c^{\gamma-1}$$

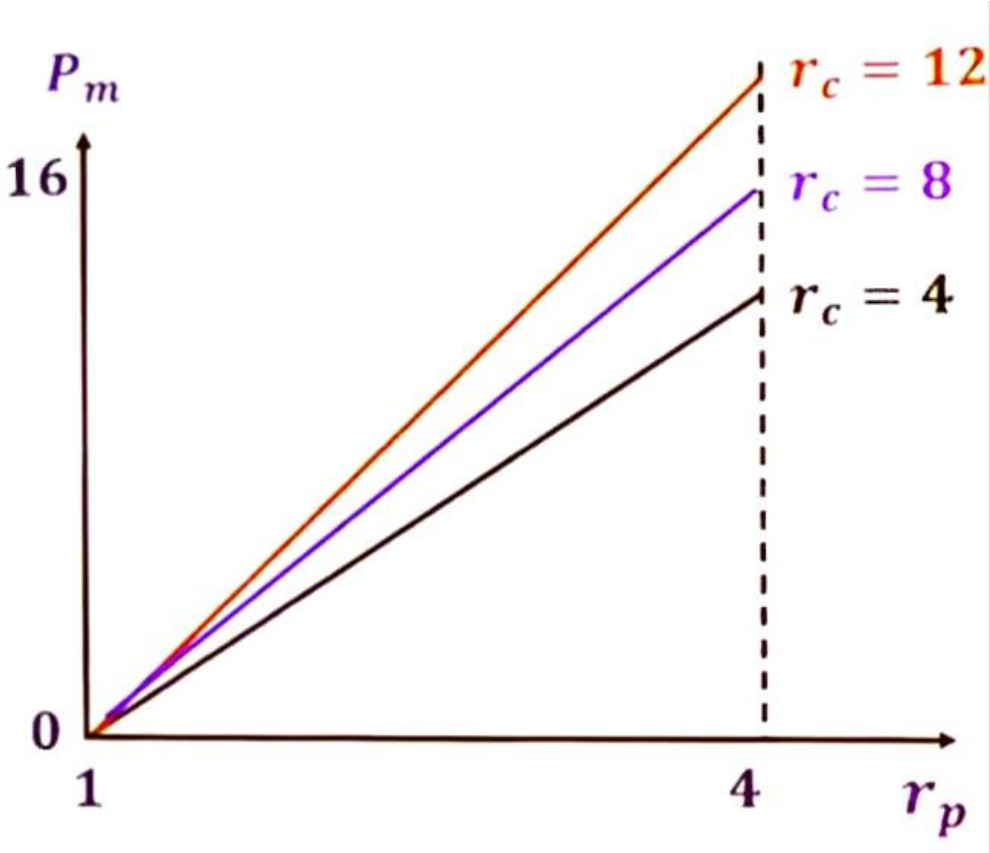
$$T_4 = r_p T_1$$

$$P_m = \frac{P_1(r_p - 1)\eta_{th} r_c^{\gamma}}{(\gamma - 1)(r_c - 1)}$$

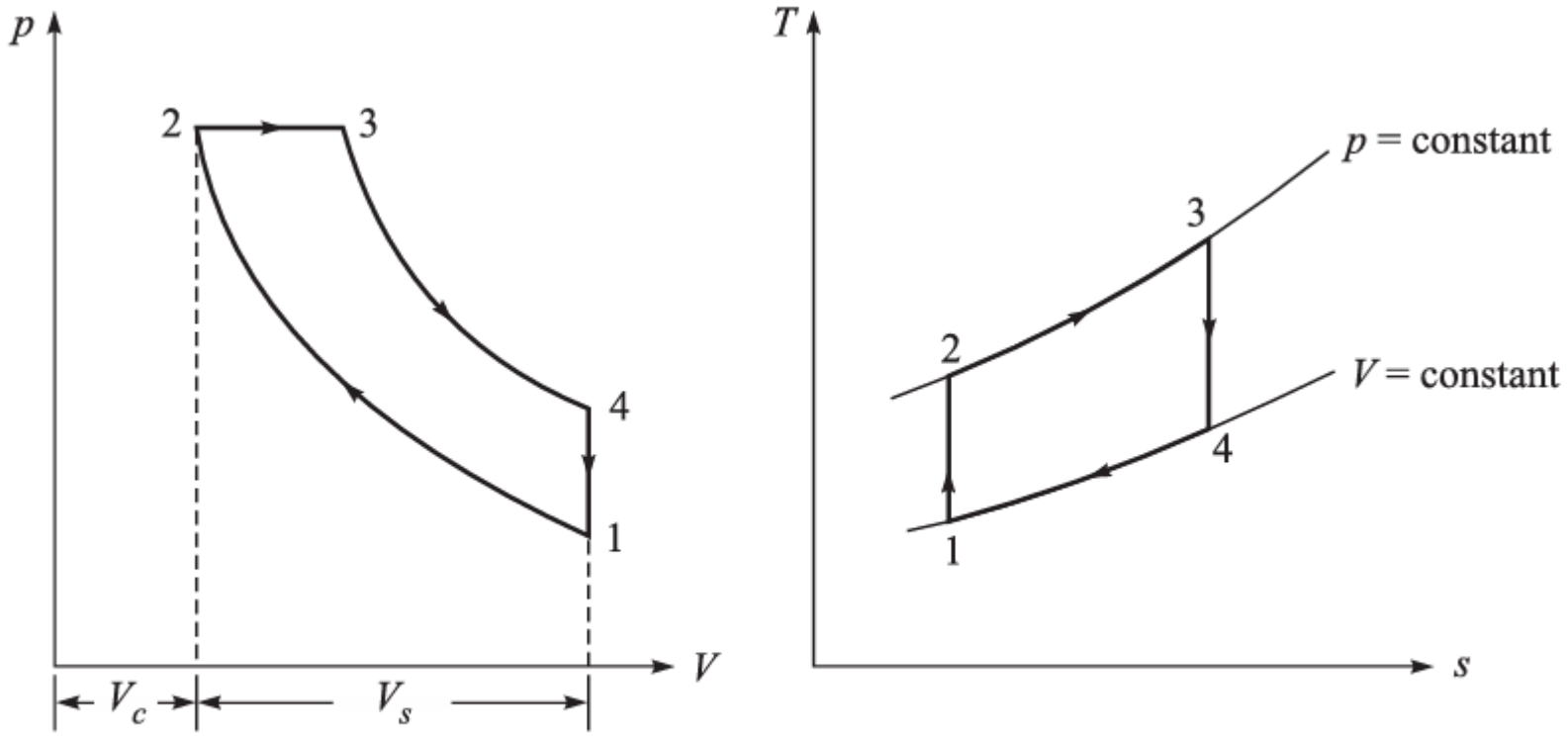
Prove

Pressure ratio

$$r_P = \frac{P_3}{P_2}$$



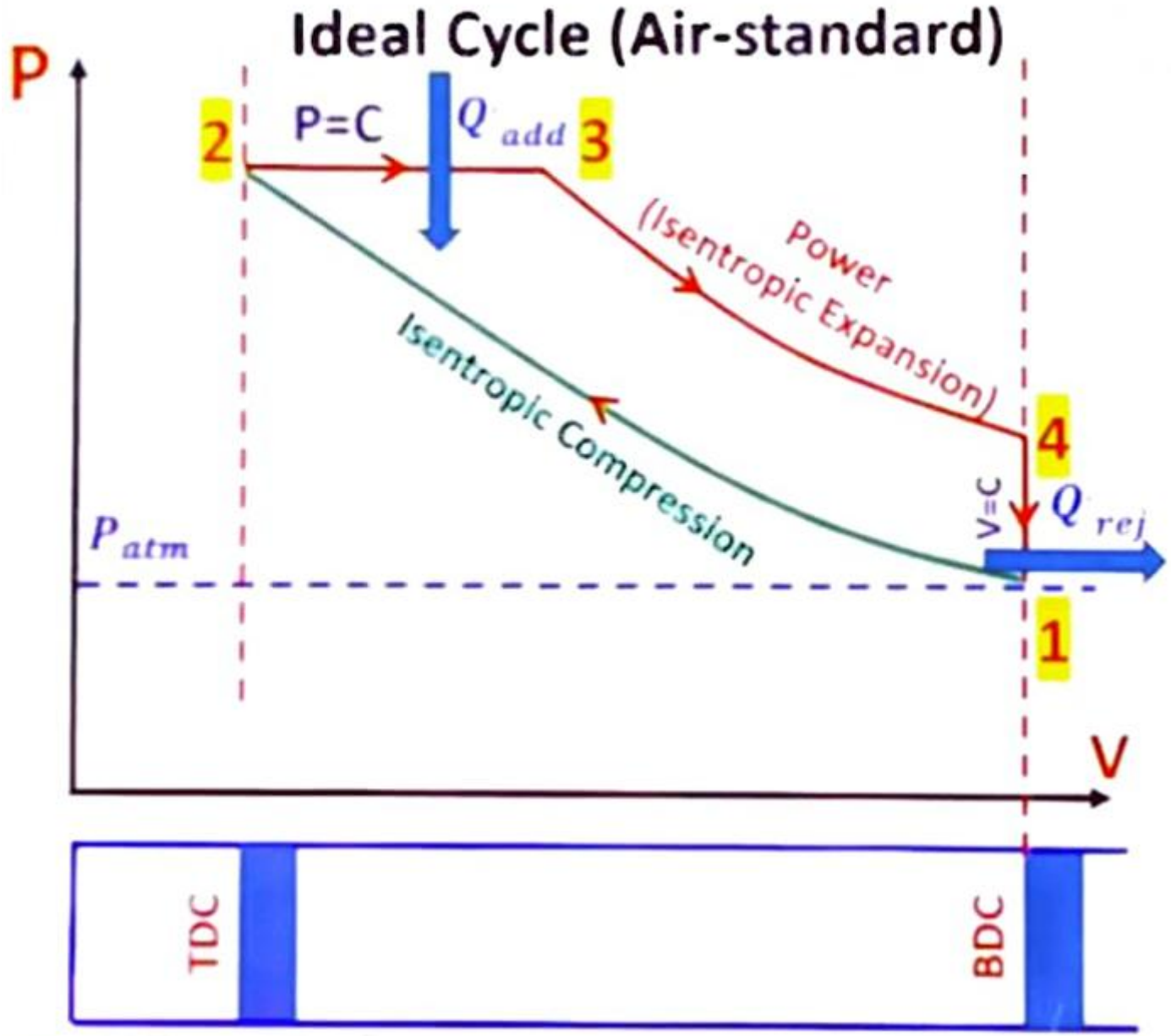
Diesel cycle



(a) $p - V$ diagram (b) $T - s$ diagram

Figure 2.11 Diesel cycle.

Process 1–2 is isentropic compression. There is no heat transfer.
Process 2–3 is reversible constant pressure process. Heat is supplied during this process.
Process 3–4 is isentropic expansion. There is no heat transfer.
Process 4–1 is reversible constant volume process. Heat is rejected during this process.



Compression ratio

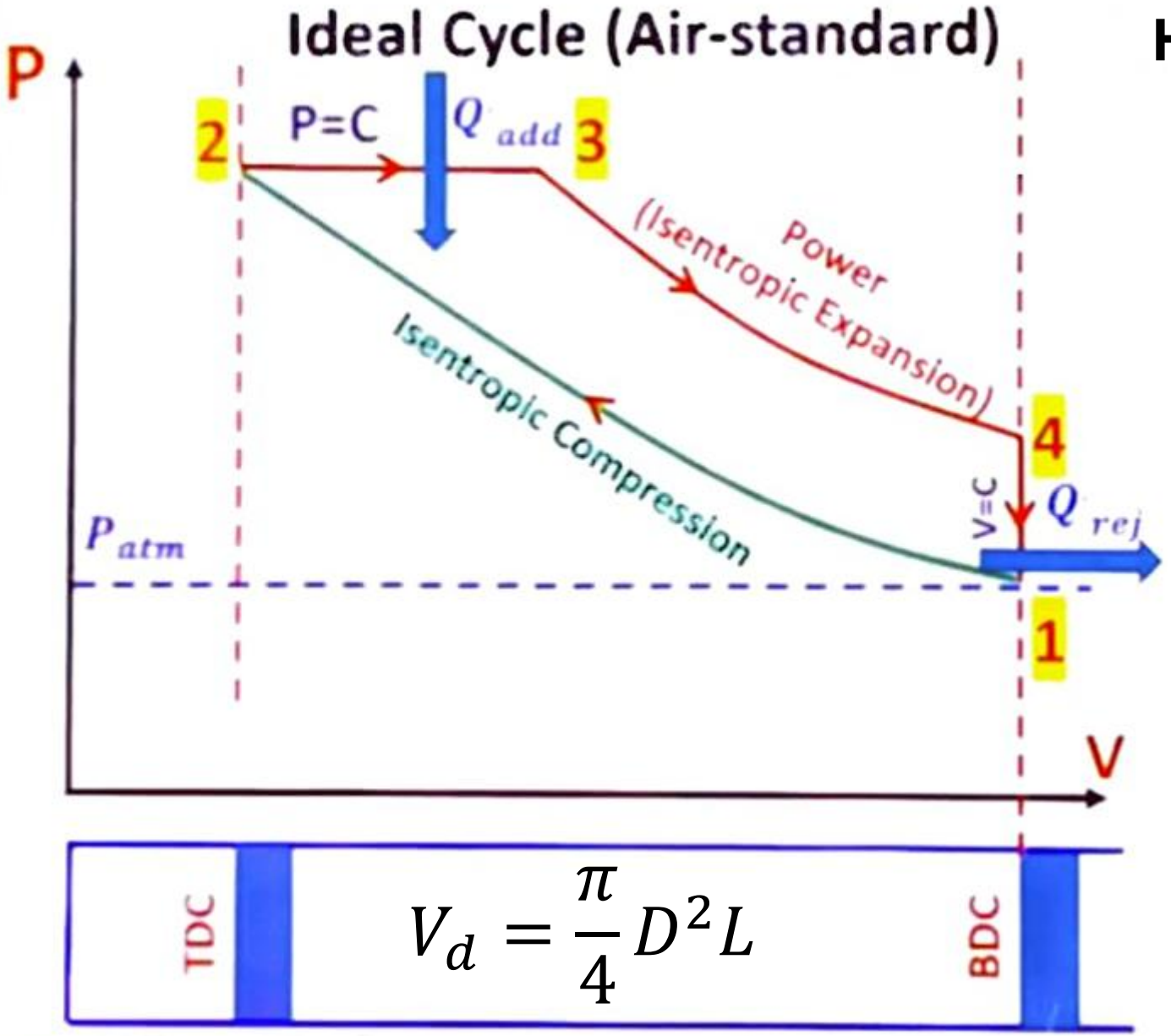
$$r_c = \frac{V_1}{V_2}$$

Cutt off Ratio

$$\beta = \frac{V_3}{V_2}$$

Swept Volume

$$V_s = V_d = \frac{\pi}{4} D^2 L$$



Heat added

$$Q_{add} = m_m c_p (T_3 - T_2)$$

$$q_{add} = c_p (T_3 - T_2)$$

Heat rejected

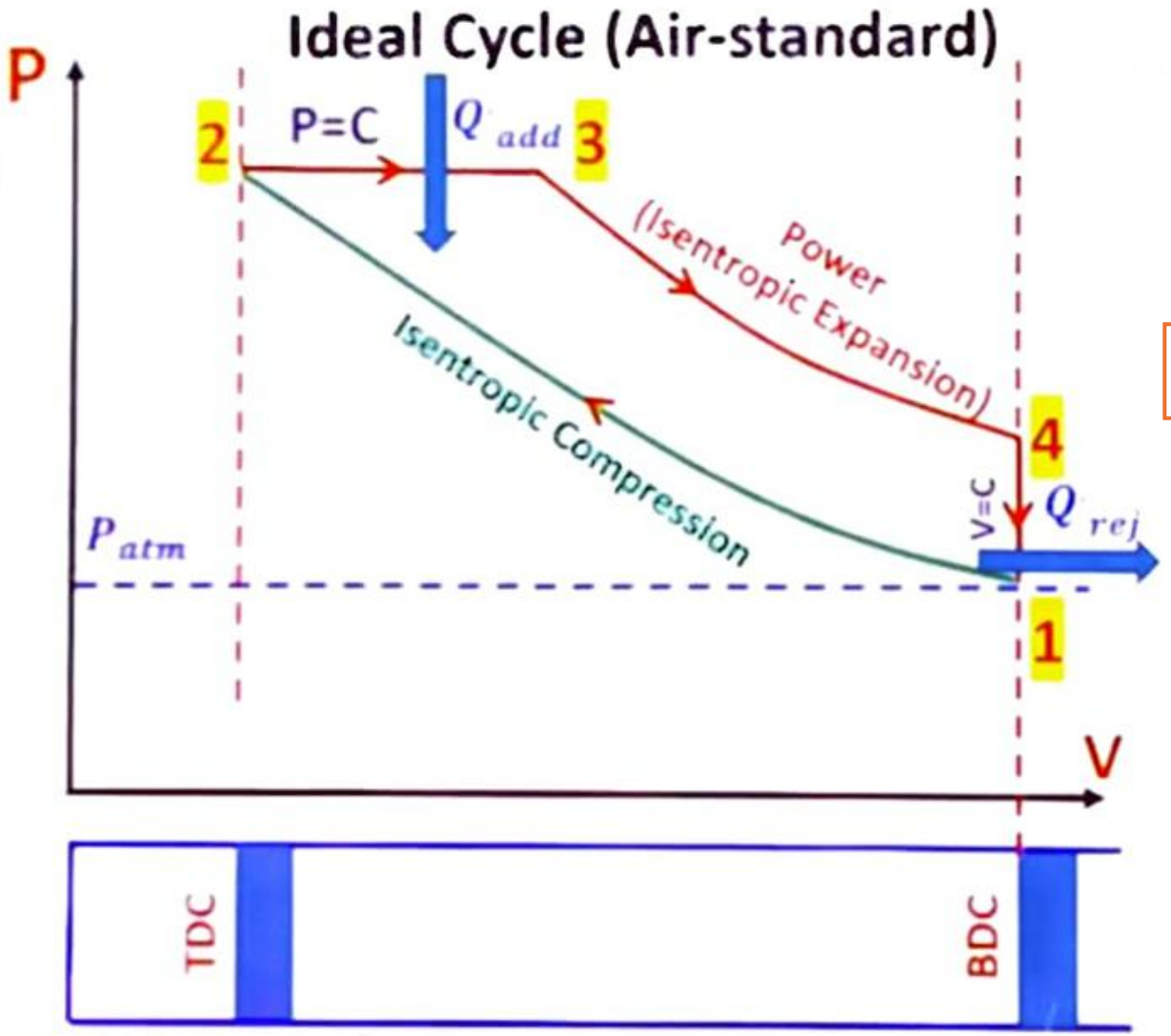
$$Q_{rej} = m_m c_v (T_4 - T_1)$$

$$q_{rej} = c_v (T_4 - T_1)$$

Compression work

$$W_{comp} = m_m c_v (T_2 - T_1)$$

$$w_{comp} = c_v (T_2 - T_1)$$



Expansion work

$$W_{1-2} = P_2(V_3 - V_2)$$

$$W_{3-4} = m\, c_v(T_3 - T_4)$$

$$\boldsymbol{W_{exp} = P_2(V_3 - V_2) + m\, c_v(T_3 - T_4)}$$

$$\boldsymbol{w_{exp} = P_2(v_3 - v_2) + c_v(T_3 - T_4)}$$

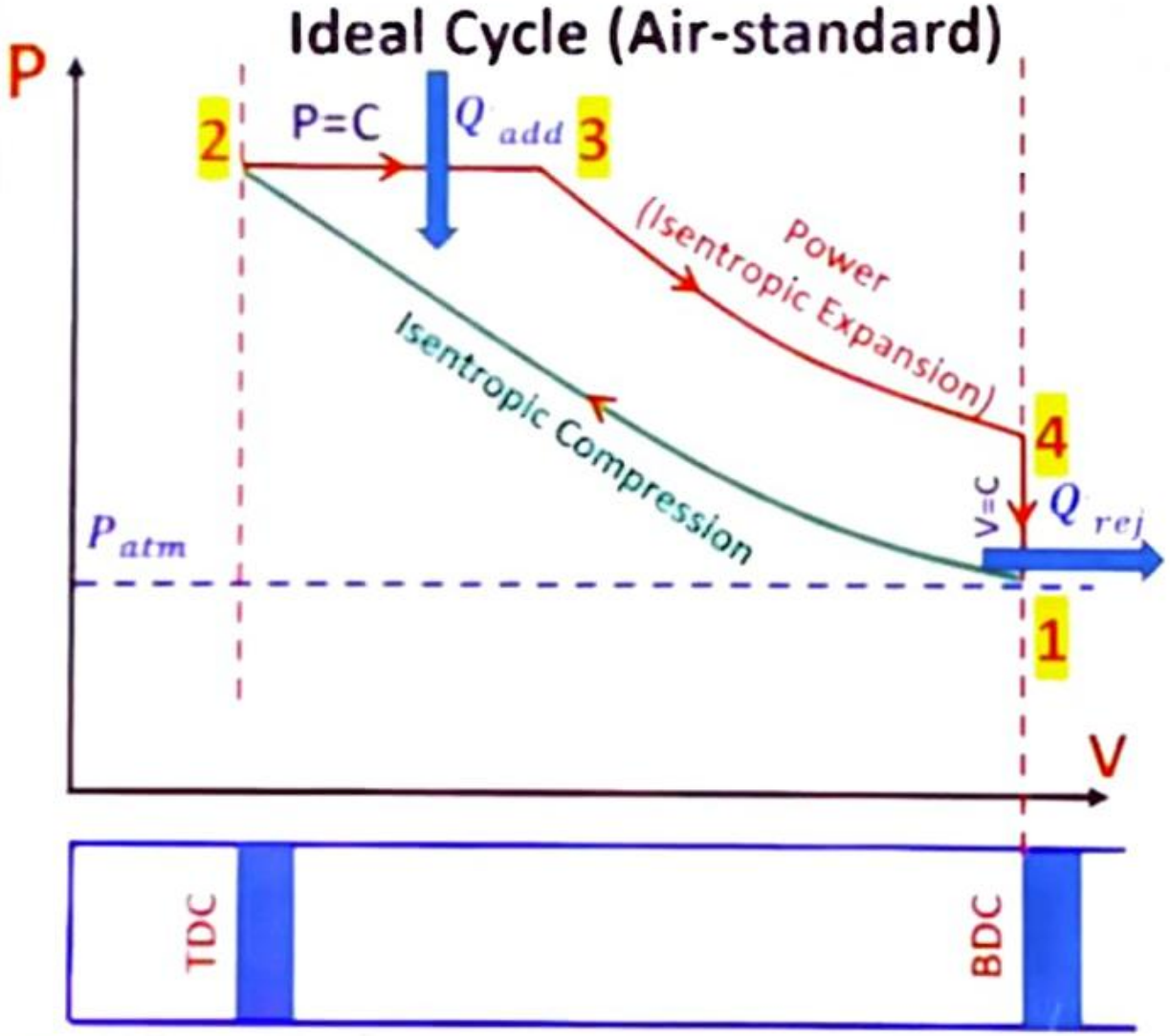
Net work

$$W_{net} = W_{comp} - W_{exp}$$

$$W_{net} = Q_{add} - Q_{rej}$$

$$w_{net} = q_{add} - q_{rej}$$

Otto Cycle analysis



Process 1-2
Isentropic compression

$$P_1 v_1 = RT_1$$

$$r_c = \frac{v_1}{v_2}$$

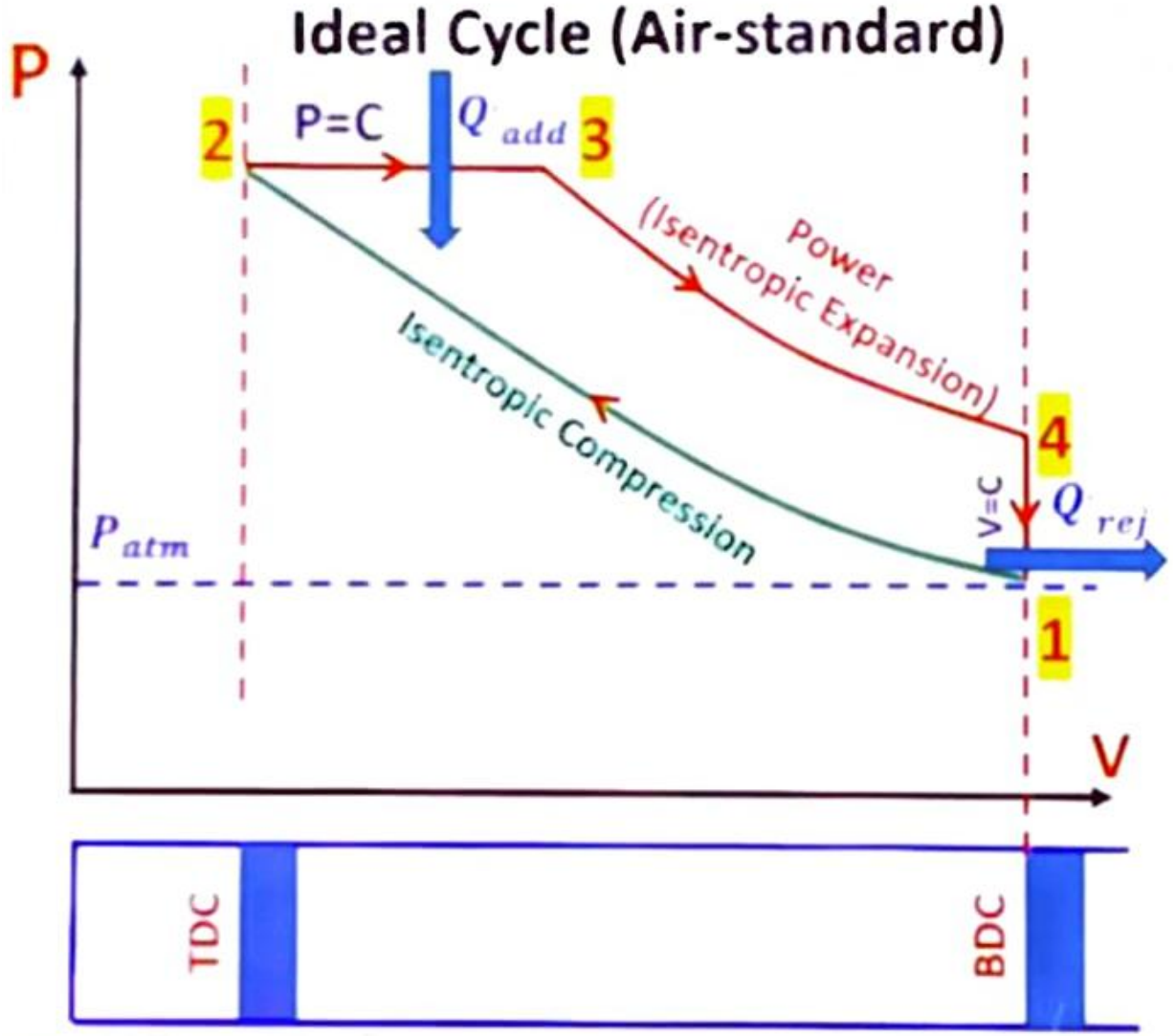
$$P_1, T_1, v_1, r_c \text{---} P_2, T_2, v_2$$

$\gamma = 1.4$ for air

$$\frac{T_2}{T_1} = (r_c)^{\gamma-1} \qquad T_2 = T_1\ (r_c)^{\gamma-1}$$

$$\frac{P_2}{P_1} = (r_c)^{\gamma} \qquad \frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$$

Otto Cycle analysis



Process 2-3
Constant pressure heat addition

$$P = const$$

$$P_2 = P_3$$

$$P_3 v_3 = R T_3$$

<u>3 Possible solutions</u>

T_{max}

β

HV or A/F

Process 2-3
Constant pressure heat addition

$$\beta = \frac{v_3}{v_2} = \frac{T_3}{T_2}$$

$$T_3 = \beta T_2$$

$$T_3 = T_1\ \beta\ (r_c)^{\gamma-1}$$

$$\frac{T_3}{T_4} = \left(\frac{v_4}{v_3}\right)^{\gamma-1} = \left(\frac{v_4}{v_2} \times \frac{v_2}{v_3}\right)^{\gamma-1}$$

$$\frac{T_3}{T_4} = \left(\frac{r_c}{\beta}\right)^{\gamma-1}$$

$$T_4 = T_3 \left(\frac{\beta}{r_c}\right)^{\gamma-1}$$

$$T_4 = T_1\ (\beta)^{\gamma}$$

Process 3-4
Isentropic Expansion

$$\frac{T_3}{T_4} = \left(\frac{v_4}{v_3}\right)^{\gamma-1}$$

$$\frac{T_3}{T_4} = \left(\frac{P_3}{P_4}\right)^{\frac{\gamma-1}{\gamma}}$$

$$\frac{P_3}{P_4} = \left(\frac{v_4}{v_3}\right)^{\gamma}$$

$$r_c = \frac{v_1}{v_2} = \frac{v_4}{v_2}$$

Thermal efficiency of Diesel cycle

$$\eta_{th} = \frac{W_{net}}{Q_{add}} = \frac{w_{net}}{q_{add}} = \frac{Q_{add} - Q_{rej}}{Q_{add}} = 1 - \frac{Q_{rej}}{Q_{add}}$$

$$\eta_{th} = 1 - \frac{\cancel{m}_m c_v (T_4 - T_1)}{\cancel{m}_m c_p (T_3 - T_2)} = 1 - \frac{(T_4 - T_1)}{k_a (T_3 - T_2)}$$

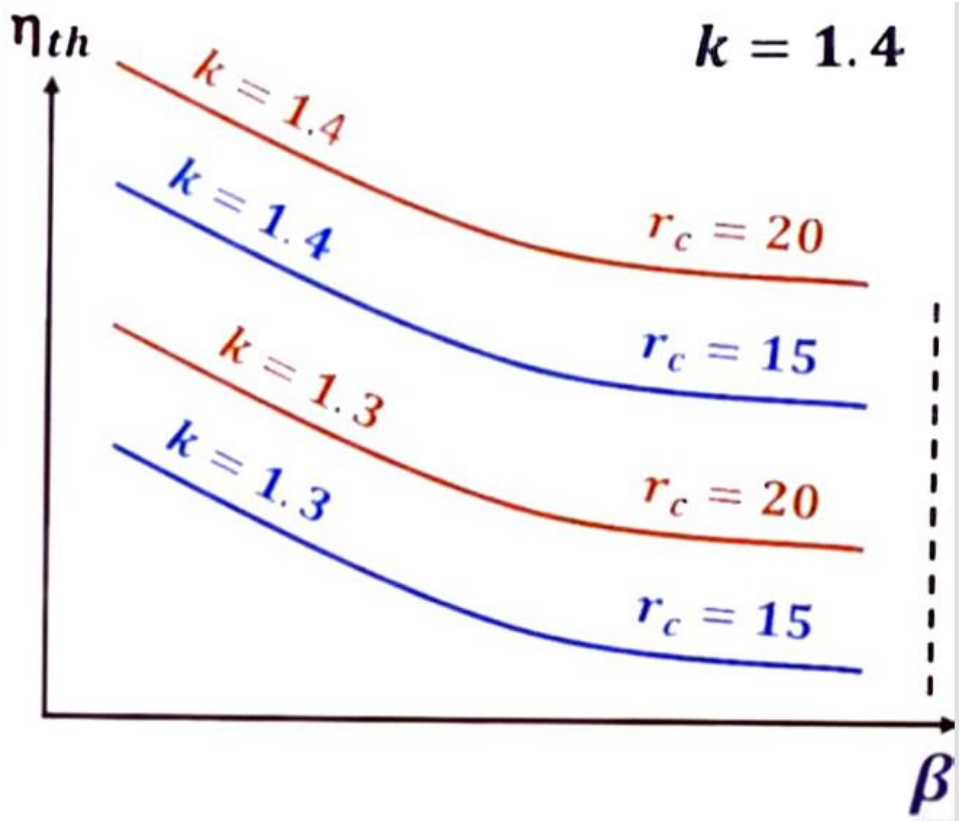
$$\eta_{th} = 1 - \frac{\cancel{(T_1(\beta)^{k_a} - T_1)}}{k_a (\cancel{T_1(\beta)(r_c)^{k_a-1}} - T_1(r_c)^{k_a-1})}$$

$$\eta_{th} = 1 - \frac{1}{(r_c)^{k_a-1}} \left(\frac{(\beta)^{k_a} - 1}{k_a(\beta - 1)} \right)$$

$$T_2 = T_1(r_c)^{k_a-1}$$

$$T_3 = \beta T_1(r_c)^{k_a-1}$$

$$T_4 = T_1(\beta)^{k_a}$$



Mean effective pressure

$$P_m = \frac{W_{net}}{V_d}$$

$$r_c = \frac{v_1}{v_2}$$

$$W_{net} = m_m\left[c_p(T_3 - T_2) - c_v(T_4 - T_1)\right]$$

$$V_d = V_1 - V_2$$

$$T_2 = T_1 r_c^{\gamma-1}$$

$$T_3 = T_1\ \beta\ (r_c)^{\gamma-1}$$

$$V_d = V_1 - \frac{v_1}{r_c}$$

$$T_4 = T_1\ (\beta)^{\gamma}$$

$$V_d = \frac{v_{1\,(r_c-1)}}{r_c}$$

$$P_m = \frac{P_1 r_c^{\gamma}}{(\gamma-1)(r_c-1)}\ \gamma(\beta - 1)\ \eta_{th}$$

Prove

EXAMPLE 2.8 In an air-standard Diesel cycle, the compression ratio is 16, the cylinder bore is 200 mm and the stroke is 300 mm. Compression begins at 1 bar and 27°C. The cut-off takes place at 8 per cent of the stroke.

Determine:

(a) The pressure, the volume and the temperature at all salient points.
(b) The cut-off ratio
(c) The work done per cycle
(d) The air-standard efficiency
(e) The mean effective pressure